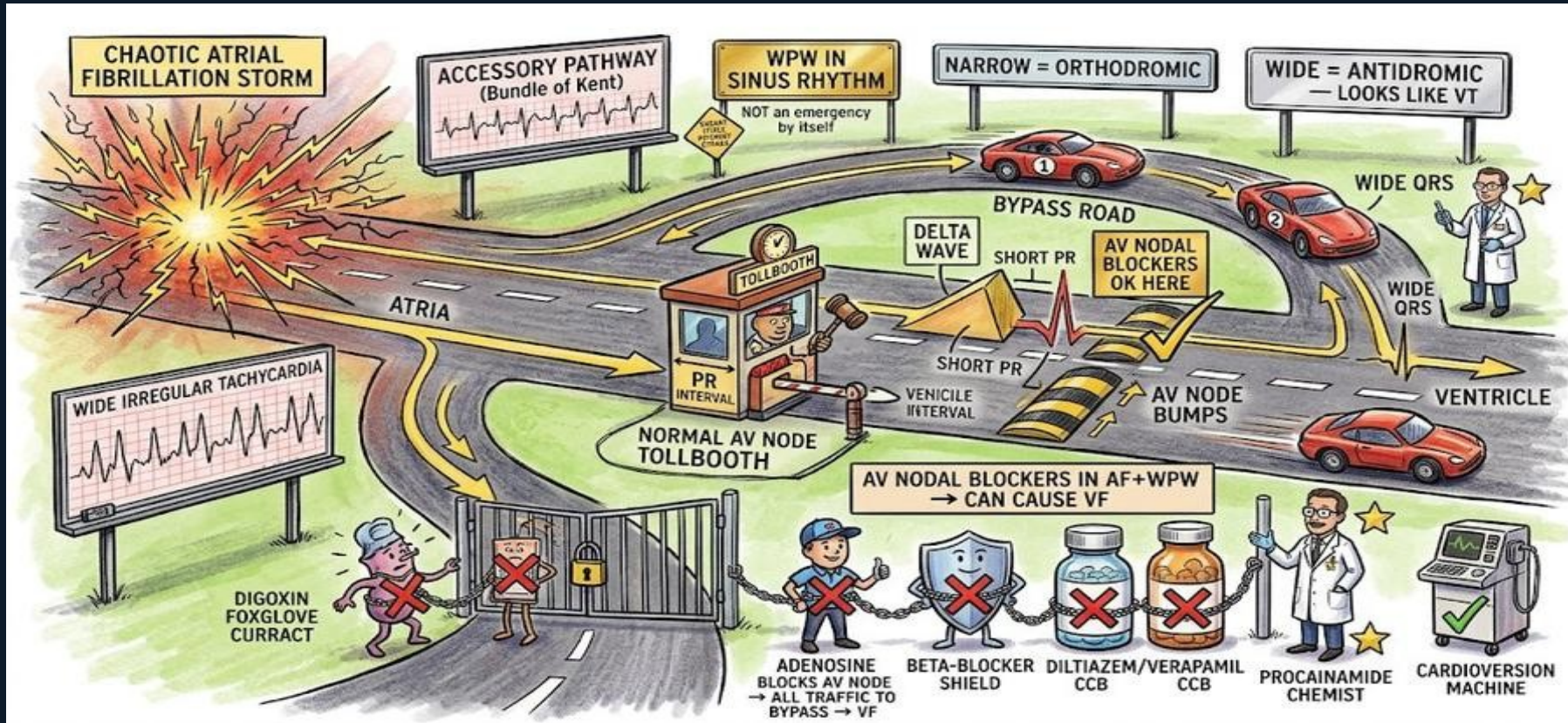
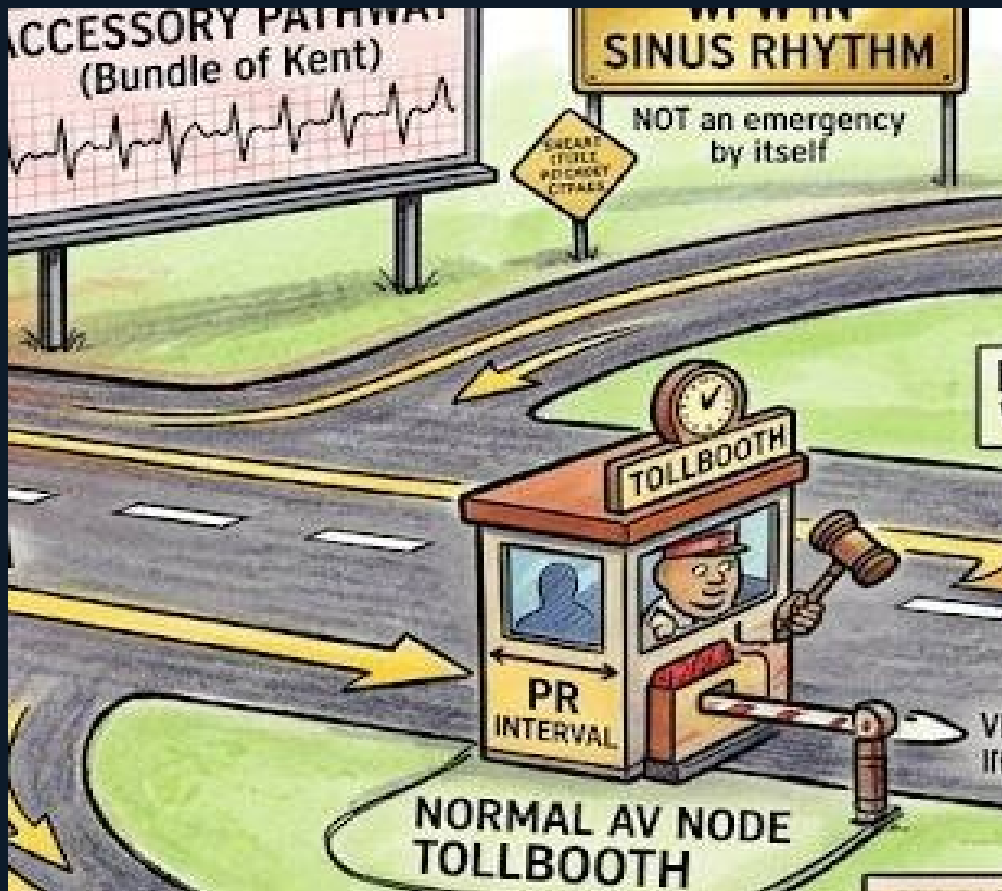


Scene F: WPW & AVRT



Electrical Highway • IMPORTANT • 2 Board Questions



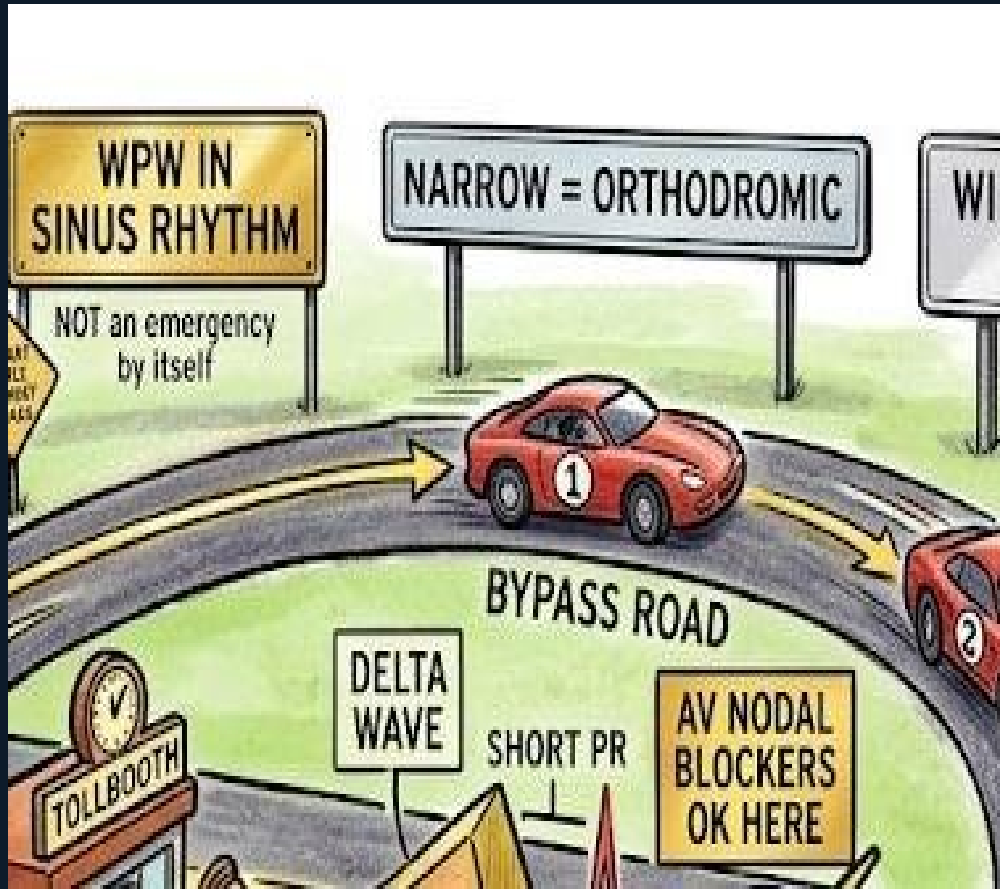
Scene F — WPW & AVRT | Arrhythmias Board Prep

The AV Node Tollbooth

The main highway passes through an AV node tollbooth where a gavel controls the PR interval — the normal gate for electrical conduction. In WPW, a bypass road above this highway circumvents the tollbooth entirely.

The delta wave (slanted ramp visible before the QRS) and short PR interval are the hallmarks of WPW in sinus rhythm — the accessory pathway pre-excites the ventricles before the normal AV node impulse arrives.

Board trap: 'WPW in sinus rhythm requires immediate treatment' — wrong. Incidental WPW without symptoms or arrhythmia = no acute treatment. Ablation considered for symptomatic patients or those with high-risk features.



Scene F — WPW & AVRT | Arrhythmias Board Prep

Orthodromic AVRT — Narrow Complex

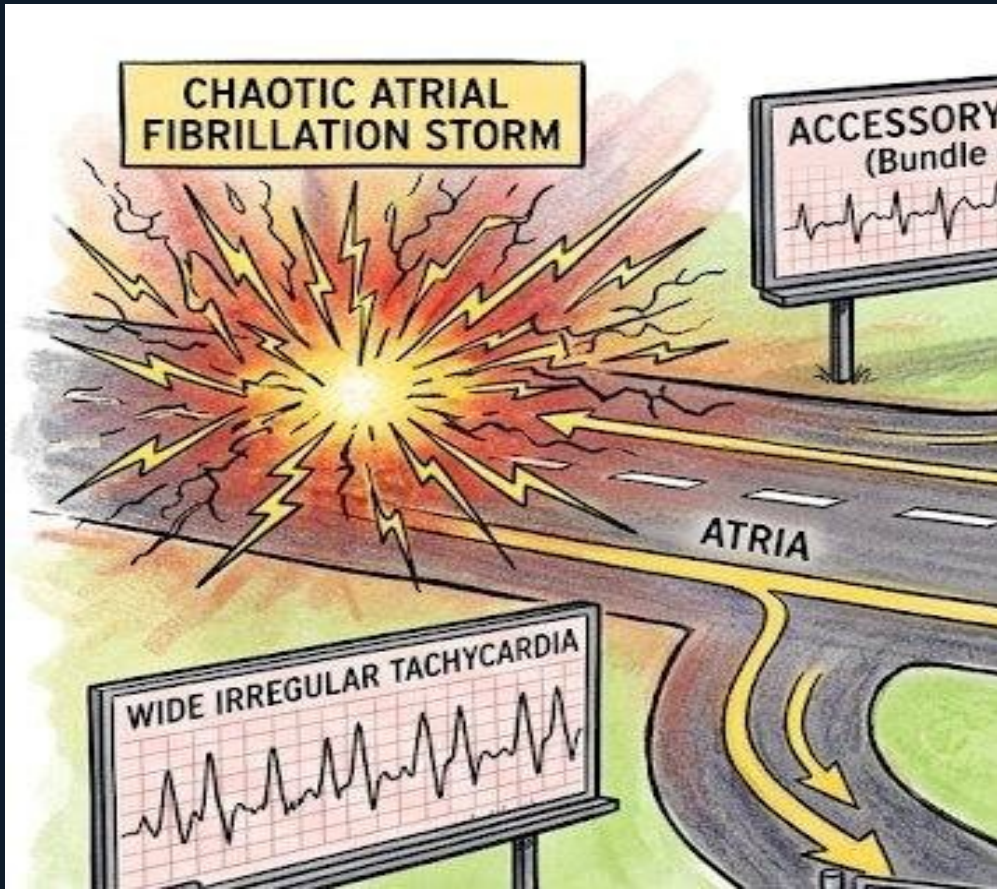
Car 1 travels DOWN the AV node, then BACK UP the accessory pathway — producing a narrow QRS. Label: 'NARROW = ORTHODROMIC.' Speed bumps on the AV node road have a checkmark: 'AV NODAL BLOCKERS OK HERE.'

Orthodromic AVRT: the reentry circuit uses the AV node antegrade (forward) and the accessory pathway retrograde (backward). Because the ventricle is activated normally through the His-Purkinje system, the QRS is narrow.

Treatment: vagal maneuvers → adenosine → AV nodal blockers (verapamil, beta-blockers). These are safe here because there is no pre-excited AF.

Car 2 travels DOWN the accessory pathway, then BACK UP the AV node — producing a wide QRS. Label: 'WIDE = ANTIDROMIC — LOOKS LIKE VT.' A gold-star procainamide chemist stands ready.

Treatment: procainamide (stable) or synchronized cardioversion (unstable). Do NOT give AV nodal blockers — if pre-excited AF is also present, AV nodal blockade can accelerate accessory pathway conduction.



AF with WPW — The Dangerous Rhythm

The AF storm erupts chaotically on the left. All electrical impulses race down the accessory pathway (bypassing the AV node filter) at enormous speed. The billboard shows an irregular wide-complex tachycardia: 'WIDE IRREGULAR TACHYCARDIA — AF + WPW.'

AF with WPW is the most dangerous arrhythmia in WPW. The AV node normally limits ventricular rate in AF to ~150. In WPW, impulses bypass this filter → ventricular rates can exceed 300 bpm → hemodynamic collapse → VF.

Board trap: giving any AV nodal blocker in AF+WPW. This is the highest-stakes wrong answer in the chapter.



The Four Forbidden Drugs

Four characters chained at the gate with red Xs:

ADENOSINE, BETA-BLOCKER SHIELD,

DILTIAZEM/VERAPAMIL, DIGOXIN FOXGLOVE. Sign: 'AV NODAL BLOCKERS IN AF+WPW → CAN CAUSE VF.'

Each of these drugs blocks the AV node, which in AF+WPW forces all electrical traffic through the accessory pathway — potentially triggering ventricular rates >300 bpm → VF.

Correct treatment: PROCAINAMIDE (stable AF+WPW) — slows accessory pathway conduction.

SYNCHRONIZED CARDIOVERSION (unstable). Both procainamide and cardioversion are shown with gold stars at the gate.

Scene F — Board Trap Master Summary

WPW in sinus rhythm = not emergency

Observe if asymptomatic; ablate if symptomatic

Antidromic AVRT → wide → procainamide

Down bypass, up AV node; looks like VT

Adenosine in AF+WPW → VF

Blocked AV node → all traffic to bypass

Diltiazem/verapamil in AF+WPW → VF

CCBs forbidden; accelerate accessory pathway

Procainamide = correct Rx for stable AF+WPW

Slows accessory pathway conduction

Orthodromic AVRT → narrow → AV blockers OK

Down AV node, up bypass; treat like SVT

AF + WPW → irregular WIDE tachycardia

Most dangerous WPW rhythm; ventricular rates >300

Beta-blockers in AF+WPW → VF

Same mechanism; absolutely forbidden

Digoxin in AF+WPW → VF

Shortens accessory pathway refractory period

Cardioversion = correct Rx if unstable

Synchronized if pulse present